

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Identify security vulnerabilities in AI-generated code
- Apply secure coding practices
- Prevent SQL Injection and XSS attacks
- Improve system reliability and safety

Task 1: Input Validation

```
import re

username = input("Enter username: ")

if re.match("^[a-zA-Z0-9]+$", username):
    print("Valid username")
else:
    print("Invalid input")
```

Task 2: SQL Injection Prevention

```
import sqlite3

conn = sqlite3.connect("test.db")
cursor = conn.cursor()

username = input("Enter username: ")

cursor.execute("SELECT * FROM users WHERE name = ?", (username,))
```

Task 3: XSS Prevention

```
import html

user_input = input("Enter comment: ")
safe_output = html.escape(user_input)

print(safe_output)
```

Task 4: Security Audit

- Avoid hardcoded passwords
- Use encryption and hashing
- Validate all inputs

Task 5: Secure Serialization

```
import json

data = {"name": "Sanju"}

with open("data.json", "w") as f:
    json.dump(data, f)
```

Conclusion

AI-generated code must be tested for security vulnerabilities.

Applying secure coding practices ensures safe and reliable applications.